ICP Crisis/Herniation Quick Reference

CLINICAL SIGNS OF HERNIATION

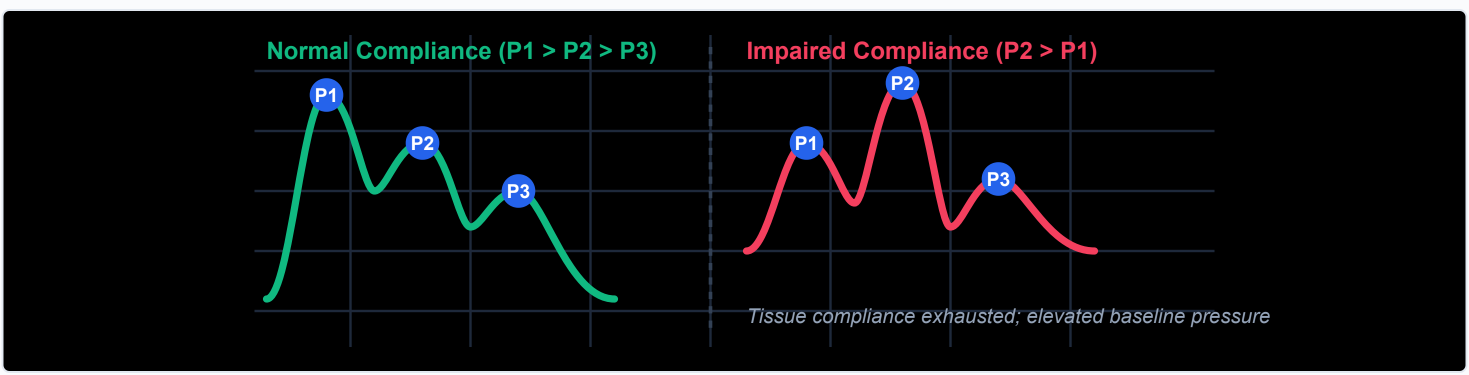
- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing's Triad is a LATE, pre-terminal sign of brainstem compression. Do not wait for its onset to initiate therapy.***

MANAGEMENT

- **Hyperosmolar Therapies & Hold Parameters:**
  - **Mannitol 20% solution:** 1 g/kg IV bolus over 20–30 min. Must use in-line 0.22-micron filter. **Hold if Serum Osmolarity > 320 mOsm/kg OR Osmolar Gap ≥ 20 mOsm/kg.**
  - **Hypertonic Saline (23.4% NaCl):** 30 mL IV bolus over 5–10 min. ***CENTRAL LINE ACCESS ONLY*** to prevent extravasation necrosis.
  - **Hypertonic Saline (3% NaCl):** 150–250 mL IV bolus over 15–20 min. Large peripheral IV access is acceptable for emergent rescue. **Hold HTS if Serum Sodium > 155–160 mEq/L or Chloride > 115–120 mEq/L.**
- **Hyperventilation:** Use strictly as short-term bridge therapy (target $PaCO_2$ 30–35 mmHg). Avoid prolonged use due to ischemia risks.
- **Steroid Contraindication:** Steroids are contraindicated for cytotoxic cerebral edema in stroke and raise infection risks.
- **Decompressive Surgery Selection Criteria:**
  - **Malignant MCA (DHC):** Age ≤ 60 years, clinical decline (GCS decline ≥ 1, pupillary changes), CT/MRI infarction ≥ 50% MCA territory, within 48h of onset (DECIMAL/DESTINY trials).
  - **Cerebellar Stroke (Suboccipital Decompression):** Mass effect on brainstem, 4th ventricle effacement, cerebellar herniation, or hydrocephalus.
- **Simplified Escalation Pathway:**

TIER 1: BASELINE	TIER 2: MEDICAL	TIER 3: BRIDGING	TIER 4: SURGICAL
hob 30°, midline neck, sedation	mannitol 1g/kg, hts 3% bolus	hts 23.4%, hyperventilation	dhc / suboccipital decomp

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**

- **Normal Compliance:** P1 > P2 > P3 (elastic brain tissue easily cushions pulsations).
- **Impaired Compliance / High ICP:** P2 > P1 (brain tissue reserve exhausted; high risk of herniation).